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Wall mount dispenser

Abstract

A wall mounted dispenser has a base. A rear mount is formed on the base and the rear mount is configured to attach to a wall. A front cover is configured to fit over the base. The modular socket is formed in the base. The pump housing is configured to fit in the modular socket as a module. The pump is mounted inside the pump housing. A liquid container bracket retains a bracket socket. The liquid container bracket is mounted to the pump housing. A nozzle bracket retains a nozzle. The nozzle bracket is mounted to the pump housing. The nozzle is connected to the pump by an exit tubing. A liquid container removably connects to the bracket socket. Liquid in the liquid container flows through the bracket socket and then to an inlet tubing. The inlet tubing is then connected to the pump.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a wall mount dispenser.

DISCUSSION OF RELATED ART

(2) A variety of bathroom soap wall mount dispensers allow dispensing of liquid cleansers, alcohol such as hand sanitizer, hand soap and body soap. A variety of different United States patents have discussed dispensing, cartridge, for example in U.S. Pat. No. 8,348,105 to Wong, entitled Compact Automatic Homogenized Liquid Detergent Dispensing Device issued Jan. 8, 2003 provides a method for pumping liquid from a soap cartridge.

(3) A counter mount dispensing system is in U.S. Pat. No. 7,752,087 to Rodenbaugh dated Jul. 13, 2010 entitled Product Dispensing System.

(4) The disclosures of the cited references are incorporated herein by reference.

SUMMARY OF THE INVENTION

(5) A wall mounted dispenser has a base. A rear mount is formed on the base and the rear mount is configured to attach to a wall. A front cover is configured to fit over the base. The modular socket is formed in the base. The pump housing is configured to fit in the modular socket as a module. The pump is mounted inside the pump housing. A liquid container bracket retains a bracket socket. The liquid container bracket is mounted to the pump housing. A nozzle bracket retains a nozzle. The nozzle bracket is mounted to the pump housing. The nozzle is connected to the pump by an exit tubing. A liquid container removably connects to the bracket socket. Liquid in the liquid container flows through the bracket socket and then to an inlet tubing. The inlet tubing is then connected to the pump.

(6) The base further includes a cover hinge, wherein the front cover is hinge mounted to the cover hinge, wherein the cover hinge is located at a lower edge of the base, wherein the cover hinge receives the front cover. The modular socket further includes a first socket retainer. The modular socket further includes a second socket retainer. The battery tray opening is formed on the pump housing. A battery tray is mounted to the battery tray opening. A motherboard can receive a controller. The controller is electrically connected to a first sensor and a second sensor. The first

sensor and the second sensor comprise a sensor assembly. The sensor assembly is mounted to the nozzle bracket. The sensor assembly is electrically connected to the controller and the sensor triggers operation of the motor.

(7) The modular socket further includes a base clip for retaining the pump housing. The base clip is formed as a leaf spring. The leaf spring has a lower engaging edge that engages the pump housing along an engagement edge. The leaf spring is configured to be snap engaged but then releasable when a user depresses the leaf spring. The base clip formed as a leaf spring is plastic injection molded integrally with the base. The pump is retained to the pump housing when the liquid container is removed.

(8) It is an object of the present invention to provide a liquid dispenser that does not require a pump connected to the liquid cartridge.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side view of the wall mount dispenser.
- (2) FIG. 2 is a front view of the wall mount dispenser.
- (3) FIG. 3 is a cross-section diagram of the wall mount dispenser.
- (4) FIG. 4 is a top view of the dispenser.
- (5) FIG. 5 is a bottom view of the wall mount dispenser.
- (6) FIG. 6 is a rear view of the wall mount dispenser.
- (7) FIG. 7 is a perspective view of the wall mount dispenser.
- (8) FIG. 8 is a perspective view of the dispenser with the cover open.
- (9) FIG. 9 is a bottom perspective view of the dispenser.
- (10) FIG. 10 is an exploded view of the dispenser.
- (11) The following call out list of elements can be a useful guide in referencing the element numbers of the drawings. **11** Key **12** Key Tooth Insert **13** Key Opening **14** Key Tooth Insert Tip **15** Latch Engagement Surface **16** Modular Socket **17** First Socket Retainer **18** Second Socket Retainer **21** Front Cover **22** Base **23** Window **24** Trim **25** Indicator Lens **26** Front Cover Latch **27** Cover Hinge **28** Rear Mount **29** Liquid Outlet **30** Liquid Container **31** Liquid Container Outlet Opening **32** Liquid Container Bracket **33** Outlet Threads **34** Liquid **35** Hollow Portion **36** Outlet Valve Gate **37** Container Adapter **38** Outlet Valve Gate Spring **39** Outlet Valve Gate Gasket **40** Pump **41** Motor **42** Inlet Pump Nipple **43** Outlet Pump Nipple **44** Bracket Socket **45** Inlet Tubing **46** Exit Tubing **47** Nozzle **48** Motor Electrical **49** Bracket Socket Tube Connector **50** Sensor Assembly **51** First Sensor **52** Second Sensor **53** Sensor Board **55** Nozzle Mount **56** Nozzle Bracket **57** Mother Board **58** Controller **59** Mounting Bracket **60** Pump Housing **61** Pump Housing Base **62** Pump Bracket **63** Pump Housing Cover **64** Battery Tray **65** Battery Tray Opening **66** Indicator Light **67** Control Button **68** Lower Bracket Mount **69** Upper Bracket Mount **70** Valve Gate Pushing Stem **71** Base Retainer **72** Leaf Spring Portion Of Base Retainer **73** Leaf Spring Lower Engaging Edge **74** Leaf Spring Slot

DETAILED DESCRIPTION OF THE EMBODIMENTS

(12) As shown in FIGS. 1-2, the side view of the dispenser and a front view of the dispenser show a base **22** having a rear mount **28** connected to a wall. The front cover **11** is connected to the base **22** at a cover hinge **27** which is at a lower edge of the base **22** and the lower edge of the front cover **21**. A window **23** is formed on the front cover **21**. A trim piece can be mounted to the front cover **21** which can provide a modular oval-shaped indicia that can receive an identifying distributor or source such as a trademark applied to the external surface of the trim piece **24**. Adjacent to the trim piece **24**, the indicator lens **25** brings light from an LED inside the front cover **21** for providing an indication to a user. A key **11** has a key tooth insert **12** which fits into a key opening **13** that triggers

a latch for opening the front cover **21**. The liquid outlet **29** which dispenses the soap is located at a lower portion of the front cover **21**. The front cover **21** can have an opening at a lower portion which allows the liquid outlet **29** to protrude from the lower portion of the front cover **21**.

(13) As seen in FIG. **3**, a cross-section of the present invention shows the key **11** with the key tooth insert **12** having a key tooth insert tip **14**. The key tooth insert tip **14** is shaped as a pointed and angled tip which engages the front cover latch **26** through the key opening **13**. The key tooth insert tip **14** abuts the latch engagement surface **15** formed on the front cover latch **26**. The front cover latch **26** is underneath the key opening **13**. The front cover latch **26** has a latch angled profile which slopes upwardly from front to back. The key tooth insert tip **14** has a tip angled profile which slopes downwardly from front to back.

(14) A liquid container **30** holds liquid **34** in a hollow portion **35** of the liquid container **30**. The liquid container **30** is underneath the front cover latch **26**. The liquid container **30** can be a flexible plastic container that fits in the front cover **21**. The liquid container **30** has outlet threads **33** that retain an outlet valve gate **36**. The outlet valve gate **36** is mounted to the liquid container outlet opening **31** and the outlet valve gate **36** controls liquid **34** flowing through the liquid container outlet opening **31**. The outlet valve gate ring **38** biases the outlet valve gate **36** to a closed position when the liquid container **30** is removed from the liquid container bracket **32**. The outlet valve gate gasket **39** seals the outlet valve gate **36** against the liquid container outlet opening **31**.

(15) The liquid container bracket **32** has a bracket socket **44**. The bracket socket receives the liquid container outlet opening **31**. The valve gate pushing stem **70** pushes the outlet valve gate **36** into open position and biases against the outlet of valve gate spring **38**. The valve gate pushing stem **70** is hollow and connects to the inlet tubing **45**. The liquid passes through the valve gate pushing stem **70** and into the inlet tubing **45** after a user loads the liquid container **30** into the liquid container bracket **32**. The inlet tubing passes to a pump **40** which then pumps the liquid to an outlet tubing or exit tubing **46**. The pump **40** is mounted to a pump housing base **61**. The pump housing base **61** has a nozzle bracket **56**. The nozzle bracket **56** retains the nozzle **47**. Soap or a liquid such as alcohol can be dispensed to the user from the nozzle **47**. The pump housing **60** includes the pump housing base **61** and retains the pump **40**, liquid container bracket **32**, and the nozzle bracket **46**. The pump housing **60** modularly attaches to the base **22** at a base retainer **71**. The base retainer **71** can be formed as a base clip. The pump housing **60** also has a sensor assembly **50** for sensing the user.

(16) As seen in FIGS. **4-5**, a top view of the dispenser and a bottom view of the dispenser show that the key **11** is pressed onto the dispenser from a top portion adjacent to the base **22**. A bottom view of the dispenser shows that the liquid outlet **29** is mounted to the nozzle **47**. The nozzle may have a pair of screws that connect to the nozzle bracket **56** between the front cover **21** and the base **22**. The first sensor **51** and the second sensor **52** are preferably mounted to the nozzle bracket **56** near and adjacent to the cover hinge **27**.

(17) As seen in FIGS. **6-7**, the rear mount **28** can be formed on the base **22**. The rear mount **28** can be formed of openings that can receive screws from the front side. The mount openings are preferably located in the upper portion of the base **22**. Preferably, a middle top opening provides an initial anchor location which then allows the user to install four separate screws. In this way, the device is securely mounted to the wall. A perspective view of the present invention shows that the housing is generally a rounded rectangular prism shape.

(18) As seen in FIG. **8**, the liquid container **30** sits on the pump housing **60**. The pump housing **60** has a control button **67** for activating different functions. An indicator light **66** illuminates the indicator lens **25**. A battery tray **64** allows the user to replace the batteries without removing the pump housing **60**. Also, the liquid outlet **29** is exposed for cleaning when the key **11** opens the front cover latch **26**. To replace the pump housing module **60**, the user can remove the liquid container and **30**, then press the base retainer **71** inwardly to slide out the pump housing module **60**.

(19) As seen in FIG. **9**, the liquid outlet **29** is mounted to the nozzle **47** and has an unobstructed clearance in case the dispenser is dispensing foam.

(20) As seen in FIG. 10, a detailed exploded view of the present invention shows all the major parts of the dispenser. The liquid container **30** has outlet threads **33**. The outlet valve gate spring **38** biases the outlet valve gate **36** in a closed position against the outlet valve gate gasket **39**. The container adapter **37** aligns the outlet valve gate **36** so that it is depressed to an open position when mounted to the bracket socket **44**. The liquid container bracket **32** holds the bracket socket **44**. The bracket socket **44** has a bracket socket tube connector **49** that connects the bracket socket **44** to the inlet tubing **45**. The inlet tubing **45** connects to the pump **40** at an inlet pump nipple **42**. The pump **40** has an inlet pump nipple **42** and an outlet pump nipple **43**. The outlet pump nipple **43** is connected to the exit tubing **46**. The exit tubing **46** is connected to the nozzle **47**. The nozzle **47** has a liquid outlet **29** mounted to it.

(21) The motor **41** in the pump **40** has motor electrical contacts **48**. The motor electrical contacts are connected to the motherboard **57** and the controller **58**. The controller **58** is configured to dispense liquid by activating the motor electrical contacts **48** when the controller **58** receives an input signal from the sensor assembly **50**. The sensor assembly **50** is mounted in the nozzle mount **55**. The nozzle mount **55** has a sensor board mount **54** which receives the sensor board **53**. The sensor board mount **54** is formed as a slot with two small circular openings at the bottom of the slot. The first sensor **51** and the second sensor **52** peek through the two small circular openings at the bottom of the slot. The sensor assembly **50** including the sensor board **53** is electrically connected to the motherboard **57** which has the controller **58**.

(22) The nozzle **47** also fits in a nozzle mount **55**. The nozzle mount **55** is also formed as a slot with an opening at the bottom for the liquid outlet **29**. The upwardly extending sidewalls of the nozzle mount **55** and the upwardly extending sidewalls of the sensor board mount **54** retain the nozzle is a rigid configuration to the first sensor **51** and the second sensor **52**.

(23) The pump bracket **62** can have a concave shape with a lower mounting opening and an upper left mounting opening for connecting and retaining the pump **42** the pump housing base **61**. The pump housing base **61** has a lower bracket mount **68** and an upper bracket mount **69**. Both the lower bracket mount and the upper bracket mount can be formed as integrally plastic injection molded screw studs for receiving three screws each in a triangular configuration. The indicator light **66** can be illuminated by an LED that is mounted to the motherboard **57**.

(24) The pump housing cover **63** fits with the pump housing base **61** to enclose the pump **40**, tubing, pair of brackets and battery tray **64**. The battery tray **64** fits into a battery tray opening **65**. The battery tray opening **65** can have electrical contacts for transmitting electrical power to the controller **58**. The pump housing **60** also has a round circular opening for the bracket socket **44**.

(25) Once the pump housing **60** is completely assembled with all of the parts, it becomes a modular unit. For repair and replacement, the pump housing module can fit onto a first socket retainer **17** and a second socket retainer **18**. Optionally, a mounting bracket **59** can mount the pump housing **60** to the modular socket **16**. The modular socket **60** is formed to receive the pump housing **60**. As an added securing feature, the base retainer **71** formed as a base clip can be formed as a resilient plastic leaf spring that has a downwardly extending engaging edge that engages the pump housing **60** one the pump housing **60** snaps onto the plastic leaf spring downwardly extending engaging edge **73**. For releasing the pump housing **60**, a user can depress the base retainer **71**. Thus, the pump housing **60** can be modularly removed and replaced without tools except for the key **11** which fits into the key opening **13**.

(26) The modular socket **16** further includes a base retainer **71** such as a base clip for retaining the pump housing. The base clip is formed as a leaf spring **72**. The leaf spring has a lower engaging edge **73** that engages the pump housing along an engagement edge. The leaf spring **72** is configured to be snap engaged but then releasable when a user depresses the leaf spring. The base clip can be formed as a leaf spring **72** that is plastic injection molded and integrally formed with the base when the injected plastic is resilient and flexible. The leaf spring **72** can be mounted in a leaf spring slot **74** that is formed on the base **22**.

(27) The liquid container is preferably disposable. The liquid container is not connected to the pump. The liquid container is pumpless and provides foam liquid.

Claims

1. A wall mounted dispenser comprising: a. a base; b. a rear mount formed on the base, wherein the rear mount is configured to attach to a wall; c. a front cover, wherein the front cover is configured to fit over the base; d. a modular socket, wherein the modular socket is formed in the base, wherein the modular socket includes a base clip configured to retain a pump housing; e. a pump housing, wherein the pump housing is configured to fit in the modular socket as a module, and wherein the base clip comprises a resilient leaf spring integrally molded with the base and having a lower engaging edge that snap-engages a mating engagement edge of the pump housing to releasably retain the pump housing in the modular socket; f. a pump powered by a motor, wherein the pump is mounted inside the pump housing; g. a liquid container, wherein the liquid container has outlet threads that retain an outlet valve gate, wherein the outlet valve gate is mounted to a liquid container outlet opening, wherein the outlet valve gate controls liquid flow through the liquid container outlet opening, wherein an outlet valve gate spring biases the outlet valve gate to a closed position when the liquid container is removed, wherein an outlet valve gate gasket seals the outlet valve gate against the liquid container outlet opening; h. a container adapter, wherein the container adapter aligns the outlet valve gate so that the outlet valve gate is depressed to an open position when mounted to a bracket socket; i. a liquid container bracket retaining a bracket socket, wherein the liquid container bracket is mounted to the pump housing, wherein the pump is retained to the pump housing when the liquid container is removed, wherein the liquid container is pumpless; j. a nozzle bracket retaining a nozzle, wherein the nozzle bracket is mounted to the pump housing, wherein the nozzle is connected to the pump by an exit tubing; and k. a liquid container removably connects to the bracket socket, wherein liquid in the liquid container flows through the bracket socket and then to an inlet tubing, wherein the inlet tubing is then connected to the pump.
2. The wall mounted dispenser of claim 1, wherein the base further includes a cover hinge, wherein the front cover is hinge mounted to the cover hinge, wherein the cover hinge is located at a lower edge of the base, wherein the cover hinge receives the front cover.
3. The wall mounted dispenser of claim 1, wherein the modular socket further includes a first socket retainer.
4. The wall mounted dispenser of claim 1, wherein the modular socket further includes a second socket retainer.
5. The wall mounted dispenser of claim 1, further including a battery tray opening formed on the pump housing, and further including a battery tray mounted to the battery tray opening.
6. The wall mounted dispenser of claim 1, further including a motherboard with a controller, wherein the controller is electrically connected to a first sensor and a second sensor, wherein the first sensor and the second sensor comprise a sensor assembly, wherein the sensor assembly is mounted to the nozzle bracket, wherein the sensor assembly is electrically connected to the controller wherein the sensor triggers operation of the motor.
7. The wall mounted dispenser of claim 1, wherein the modular socket further includes a base clip for retaining the pump housing.
8. The wall mounted dispenser of claim 7, wherein the base clip is formed as a leaf spring.
9. The wall mounted dispenser of claim 8, wherein the leaf spring has a lower engaging edge that engages the pump housing along an engagement edge.
10. The wall mounted dispenser of claim 9, wherein the leaf spring is configured to be snap engaged but then releasable when a user depresses the leaf spring.
11. The wall mounted dispenser of claim 10, wherein the base clip formed as a leaf spring is plastic injection molded integrally with the base.

12. The wall mounted dispenser of claim 11, wherein the base further includes a cover hinge, wherein the front cover is hinge mounted to the cover hinge, wherein the cover hinge is located at a lower edge of the base, wherein the cover hinge receives the front cover.
13. The wall mounted dispenser of claim 11, wherein the modular socket further includes a first socket retainer.
14. The wall mounted dispenser of claim 11, wherein the modular socket further includes a second socket retainer.
15. The wall mounted dispenser of claim 11, further including a battery tray opening formed on the pump housing, and further including a battery tray mounted to the battery tray opening.
16. The wall mounted dispenser of claim 11, further including a motherboard with a controller, wherein the controller is electrically connected to a first sensor and a second sensor, wherein the first sensor and the second sensor comprise a sensor assembly, wherein the sensor assembly is mounted to the nozzle bracket, wherein the sensor assembly is electrically connected to the controller wherein the sensor triggers operation of the motor.
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